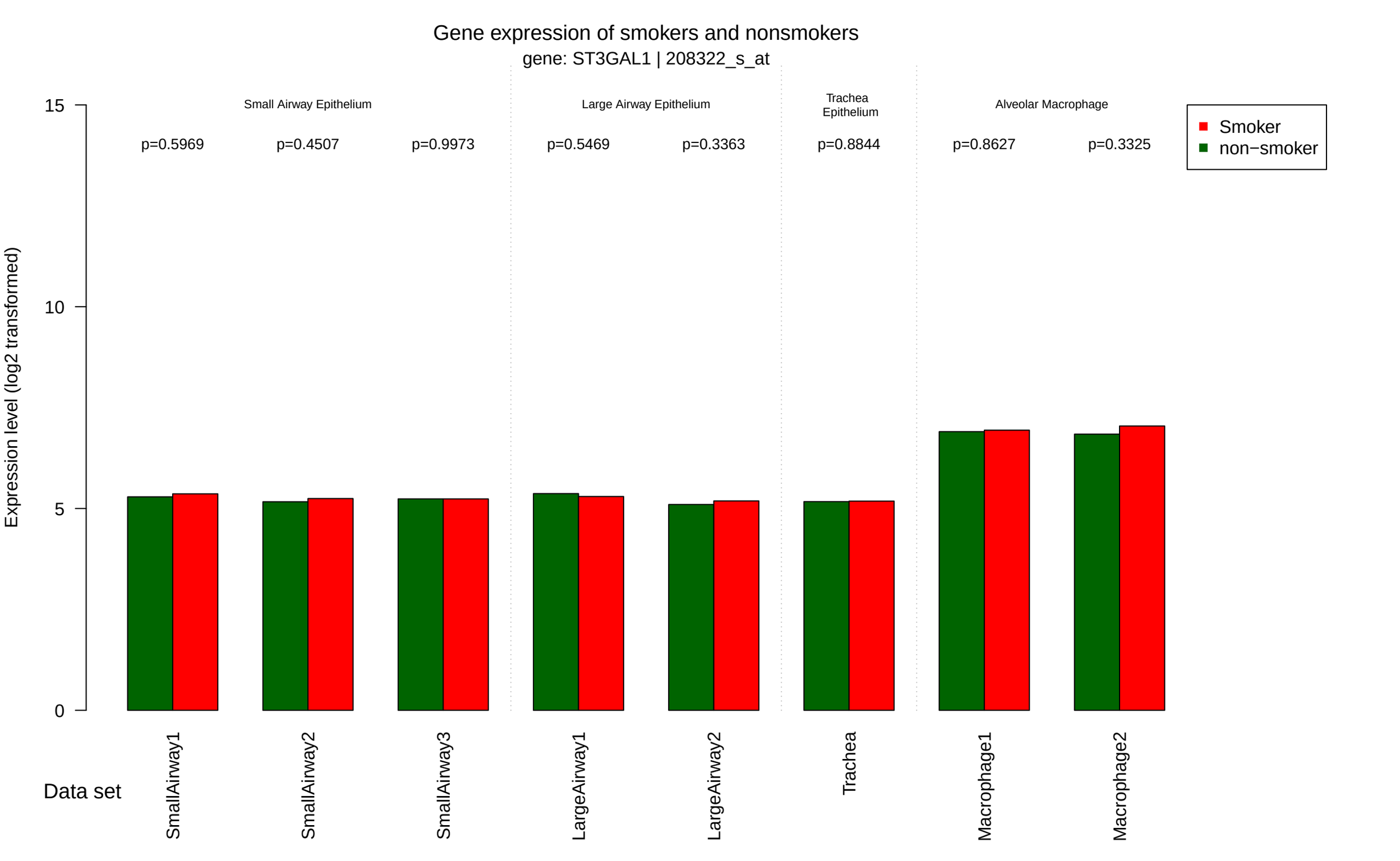
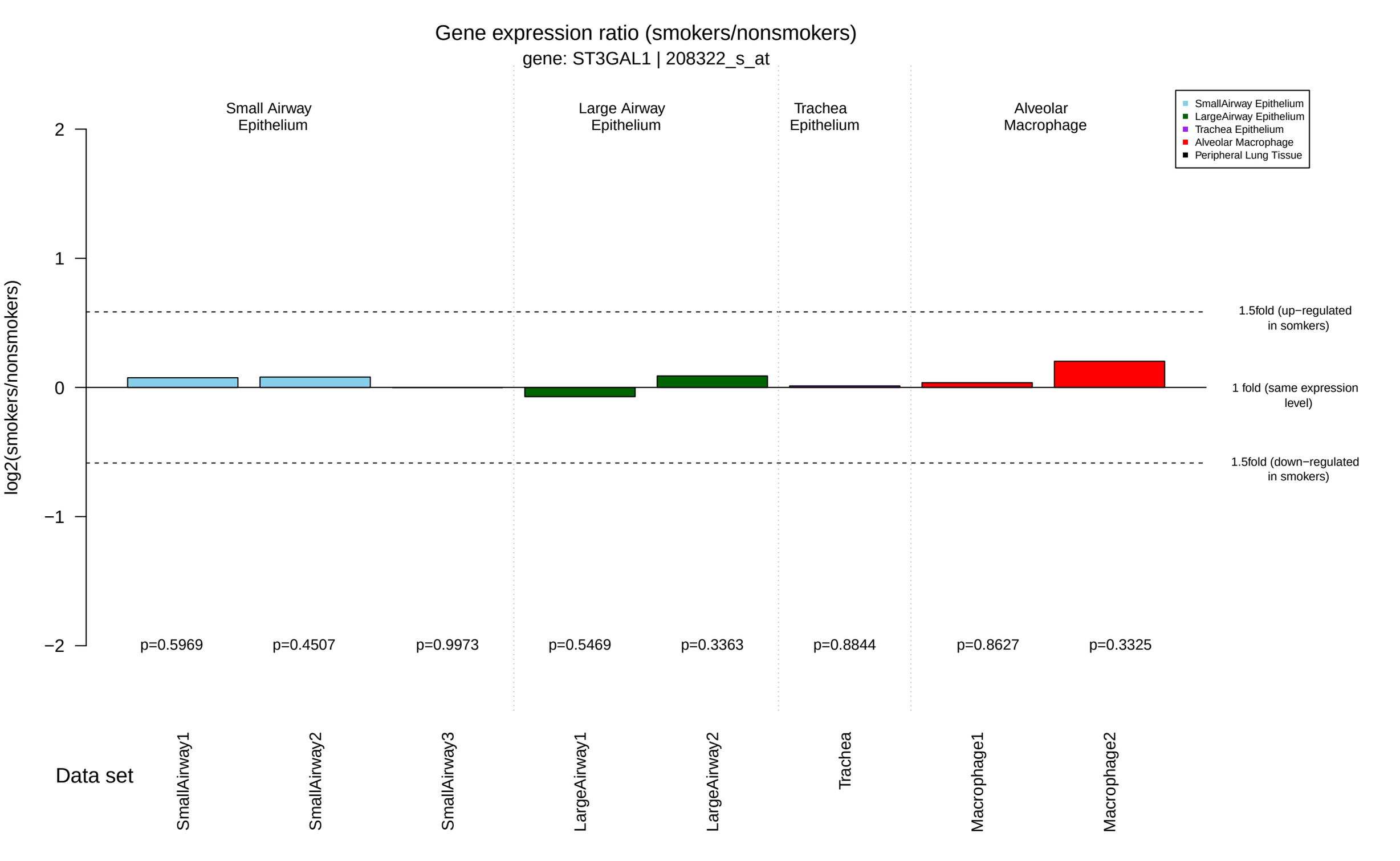
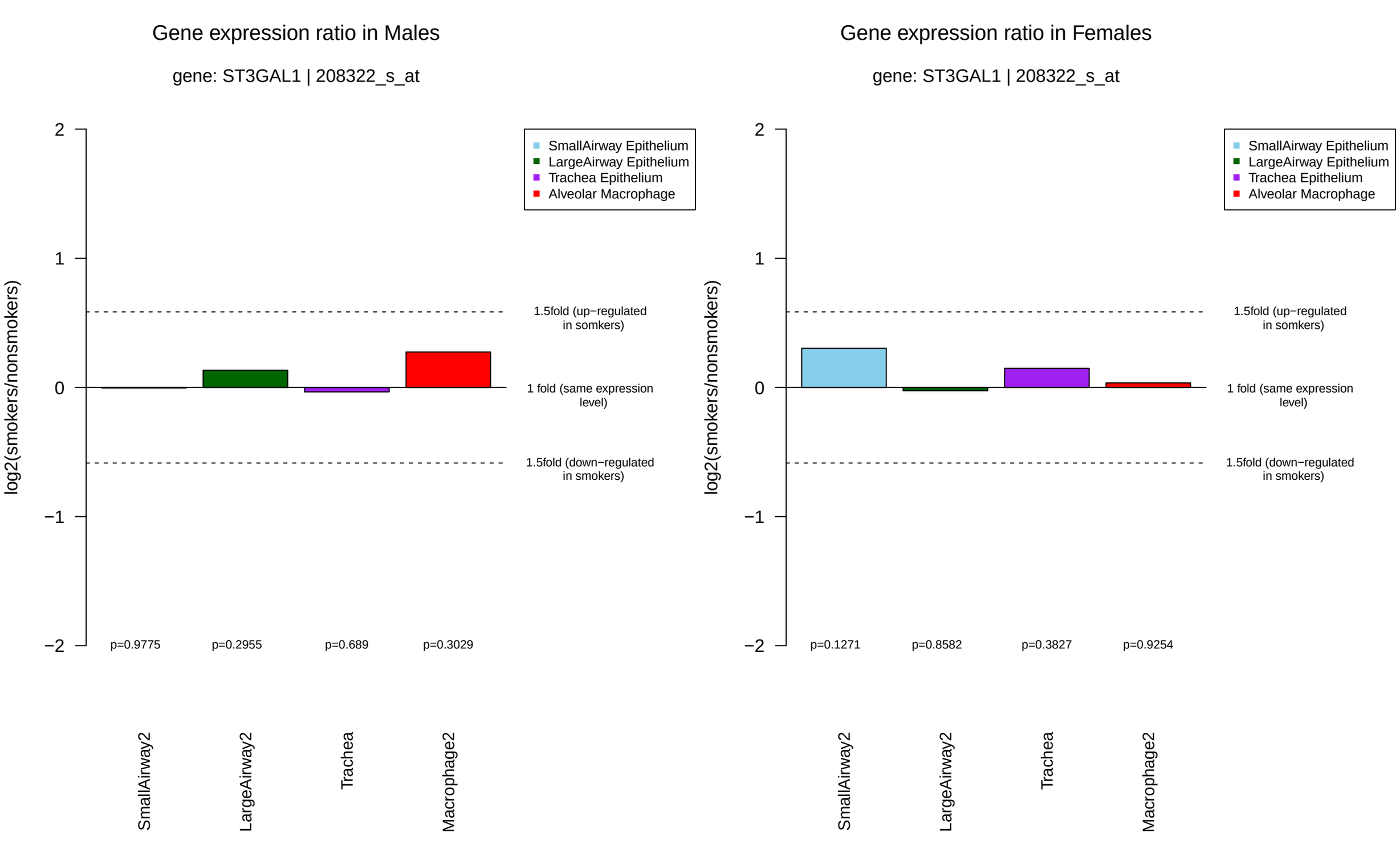
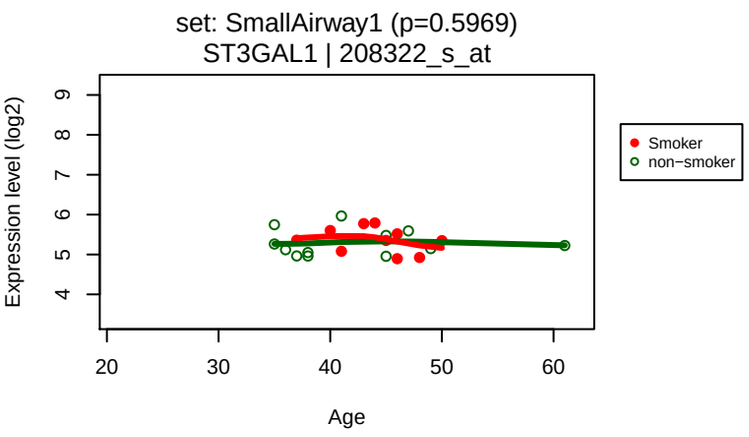
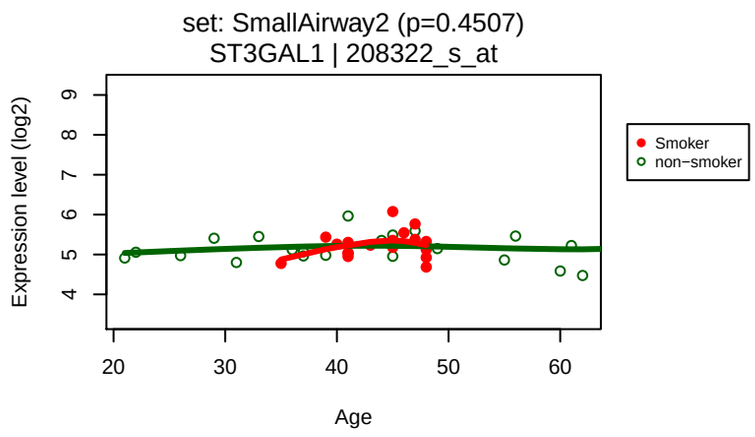
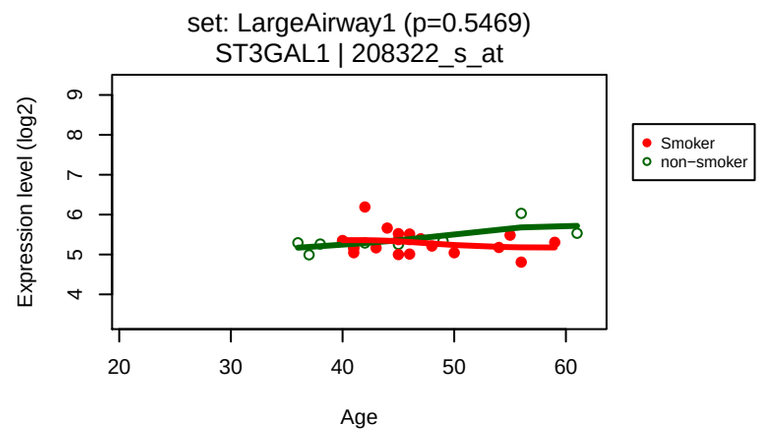

Gene expression of smokers and nonsmokers
gene: ST3GAL1 | 208322_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.5969
p=0.4507
p=0.9973
p=0.5469
p=0.3363
p=0.8844
p=0.8627
p=0.3325
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

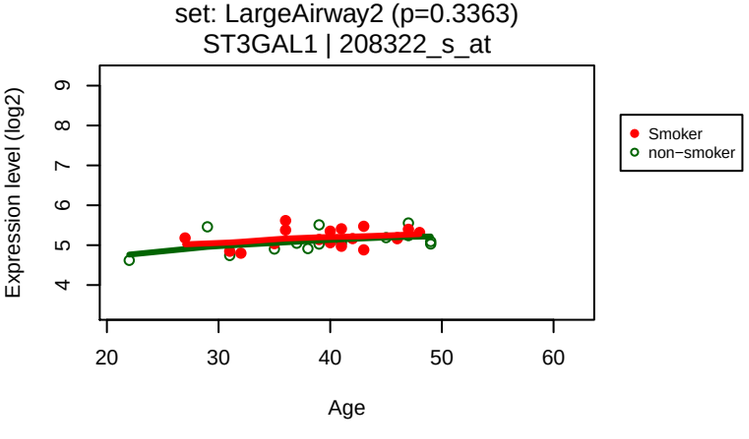

Gene expression ratio (smokers/nonsmokers)
gene: ST3GAL1 | 208322_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5969
p=0.4507
p=0.9973
p=0.5469
p=0.3363
p=0.8844
p=0.8627
p=0.3325
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

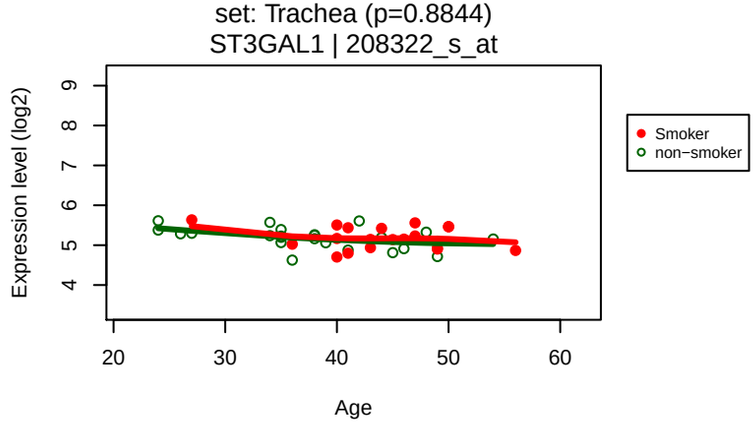

Gene expression ratio in Males
gene: ST3GAL1 | 208322_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9775
p=0.2955
p=0.689
p=0.3029
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ST3GAL1 | 208322_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1271
p=0.8582
p=0.3827
p=0.9254
SmallAirway2
LargeAirway2
Trachea
Macrophage2

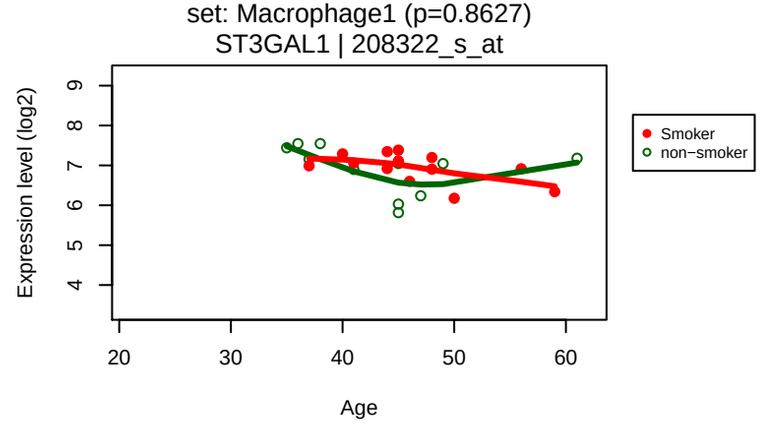

set: SmallAirway1 (p=0.5969)
ST3GAL1 | 208322_s_at
Expression level (log2)
Age
Smoker
non−smoker

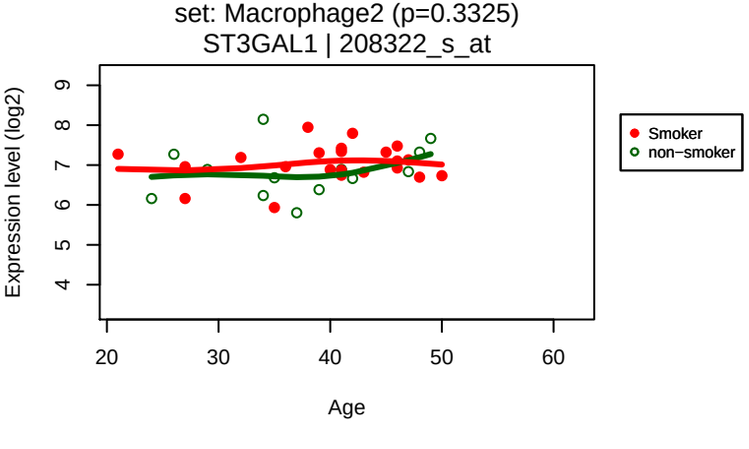

set: SmallAirway2 (p=0.4507)
ST3GAL1 | 208322_s_at
Expression level (log2)
Age
Smoker
non−smoker

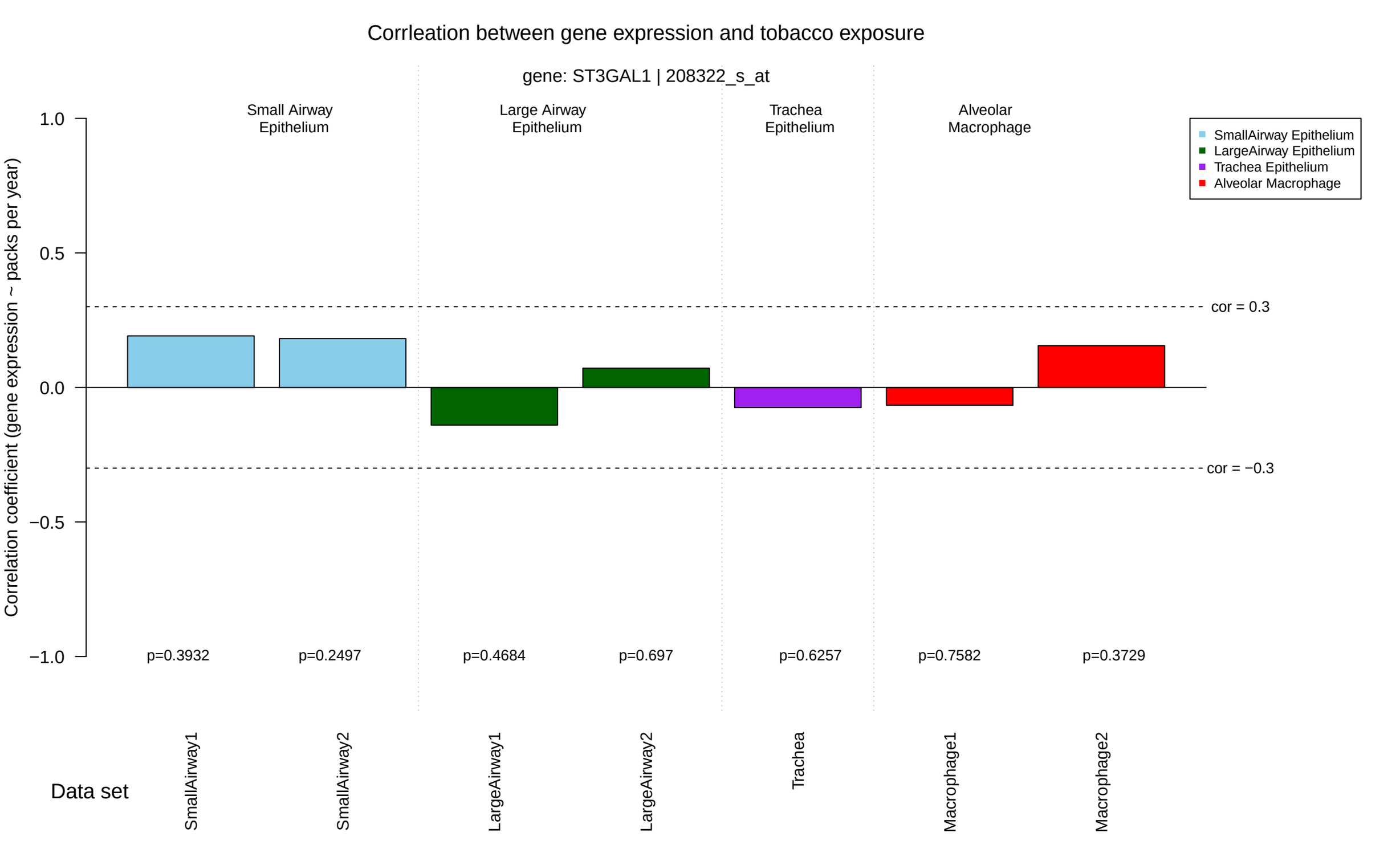

set: LargeAirway1 (p=0.5469)
ST3GAL1 | 208322_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.3363)
ST3GAL1 | 208322_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.8844)
ST3GAL1 | 208322_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.8627)
ST3GAL1 | 208322_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.3325)
ST3GAL1 | 208322_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ST3GAL1 | 208322_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
Correlation coefficient (gene expression ~ packs per year)
p=0.3932
p=0.2497
p=0.4684
p=0.697
p=0.6257
p=0.7582
p=0.3729
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2